Formula Sheet

$\Delta NII_i = (RSA_i - RSL_i)\,\Delta R_i = (GAP_i)\Delta R_i$

$\Delta NII_i = (CGAP_i)\Delta R_i$

$\Delta NII = (RSA \times \Delta R_{RSA}) - (RSL \times \Delta R_{RSL})$

$M_{GAP} = M_A - M_L$

$$PV = C\left[\frac{1-(1+i)^{-n}}{i}\right] + \frac{FV}{(1+i)^n}$$

$$D = \frac{\sum_{t=1}^{N} CF_t \times DF_t \times t}{\sum_{t=1}^{N} CF_t \times DF_t} = \frac{\sum_{t=1}^{N} PV_t \times t}{\sum_{t=1}^{N} PV_t}$$

$D_c = 1 + 1/R$

$\Delta P = -D[\Delta R/(1+R)]P = -(MD) \times (\Delta R) \times (P)$

Dollar duration = MD × Price

$$\Delta E = -(D_A - D_L k) \times A \times \frac{\Delta R}{(1+R)}$$

$$D_p = \Sigma(MV_i \times D_i) / \sum MV_i$$

DEAR = *Dollar market value of the position × daily return under adverse circumstances*

$$\mathbf{Ret} = \frac{\Delta P}{P} = -\left(\frac{D}{1+R}\right)\Delta R$$

$Ret_P = w_1 * Ret_1 + w_2 * Ret_2$

N-day VAR = DEAR $\times\sqrt{N}$

$DEAR_p = [DEAR_1^2 + DEAR_2^2 + 2\rho_{1,2} \times DEAR_1 \times DEAR_2]^{1/2}$

$DEAR_p = [DEAR_1^2 + DEAR_2^2 + DEAR_3^2 + 2\rho_{1,2} \times DEAR_1 \times DEAR_2 + 2\rho_{1,3} \times DEAR_1 \times DEAR_3 + 2\rho_{2,3} \times DEAR_2 \times DEAR_3]^{1/2}$

1+k = 1 + [of + (BR+ Φ)] / [1- b(1-RR)]

E(1+r)=1+E(r) = p(1+k)

C = k – i

1+i = p(1+k)

$$1+i = \left[(1-p)\gamma(1+k)\right] + \left[p(1+k)\right]$$

$$\mathrm{RAROC} = \frac{\text{one year income on a loan}}{\text{loan (asset) risk or capital at risk}}$$

$\Delta L = - D * L * [\Delta R / (1+R)]$

$$\text{Concentration limit} = \text{Maximum loss as a percentage of capital} \times \frac{1}{\text{Loss rate}}$$

$$\sigma_p^2 = \sum_{i=1}^{n} X_i^2 \sigma_i^2 + \sum_{i=1}^{n} \sum_{\substack{i=1 \\ i \neq j}}^{n} X_i X_j \rho_{ij} \sigma_i \sigma_j$$

$R_i = R_f + Beta_i * (R_m - R_f)$

$Beta_i = Covariance(R_i, R_m)/Variance(R_m)$

$$R_i = AIS_i \times (1 - EDF_i) + (AIS_i - LGD_i) \times EDF_i$$
$$= AIS_i - EDF_i \times LGD_i$$

$$\sigma_i = \sqrt{EDF_i (1 - EDF_i)} \times LGD_i$$

$$\sigma_j = \sqrt{\frac{\sum_{i=1}^{N} (X_{ij} - X_i)^2}{N}}$$

$l = \sum w_i (P_i / P_i^*)$